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Exact Markov Chain Monte Carlo Methods for the Normalized Power Prior

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Statement of Authorship

I hereby declare that the thesis submitted is my own work. All direct or indirect sources used are acknowledged as references. I further declare that I have not submitted this thesis at any other institution in order to obtain a degree.

To anyone who has ever believed in Education as the main way to overcome adversity.

Acknowledgements

I would like to thank [many people]

"Agora eu era o herói, e o meu cavalo só falava inglês..."

"Now, I was the hero, and my horse only spoke* in English..."*

From *João e Maria* (1977) by Chico Buarque and Sivuca.

Abstract

In statistical modelling with historical data, a central challenge is how to effectively integrate prior evidence with current observations. This issue is particularly pressing in clinical trials and medical applications, where using historical data can improve both the efficiency and robustness of inference. A natural Bayesian strategy for this purpose is the use of informative priors. Among them, the normalised power prior has attracted significant interest, as it provides a principled way to control the influence of historical information. The method introduces a power parameter, $\eta \in [0, 1]$, that governs the degree of borrowing: $\eta = 0$ discards the past data while $\eta = 1$ fully incorporates it. This flexibility makes the framework highly appealing in practice. Despite its advantages, the power prior posterior distribution is doubly intractable, as it requires evaluation of a normalising constant that depends on η. Existing approaches rely on approximate inference, which often lack theoretical guarantees and explicit convergence properties. To overcome these limitations, we propose an exact MCMC algorithm based on pseudo-marginal methods. The key idea is to replace the intractable ratio of normalising constants with an unbiased estimator. After finding useful identities for the derivative of the log normalising constant, we construct unbiased estimators of log-ratios through Monte Carlo integration, enabling exact acceptance probabilities, by employing the Poisson Estimator Method, within a Metropolis-Hastings or Barker's scheme. We evaluate the method on Gaussian models under varying degrees of compatibility between historical and current data. Results demonstrate that the algorithm accurately reflects data conflict by shrinking η when the datasets are incompatible, while recovering correct posterior behaviour when they are aligned. Comparisons with analytical baselines confirm that our method achieves reliable estimates of both model parameters and the power parameter, with efficiency suitable for practical use. This provides a rigorous, exact alternative to approximate approaches, offering stronger guarantees for applications such as adaptive clinical trial design.

Keywords: Bayesian Inference; Power Prior; Doubly Intractable Distributions; Exact MCMC; Historical Data Analysis

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1 Introduction

`Luiz:` Second-to-last thing to be written. Don't sweat it. Leave it to *Papa*

- Describe NPP and notation; motivation with historical data (MANY CITATIONS).
- The NPP as a doubly-intractable problem (the prior is intractable)
- Mention that we start with approximate methods and then move on to exact stuff

In statistical modelling with historical data, a central challenge is effectively integrating information from past studies with current data. This is particularly relevant in clinical trials and medical research, where leveraging historical data can improve the efficiency and robustness of statistical inferences. A natural approach in this setting is the use of *informative priors*, which incorporate prior knowledge into the analysis, thereby enhancing parameter estimation and decision-making. `Adame:` Some references here would be nice.

Among the various methods for constructing informative priors, *power priors* (Ibrahim et al., 2015) have gained significant attention. They allow for controlled borrowing of information from historical data by adjusting a power parameter, η, which typically ranges between 0 and 1. When $\eta = 0$, no historical information is incorporated, whereas $\eta = 1$ fully integrates the historical data into the current analysis. `Adame:` I would like to introduce the NPPs issues here and then talk about the motivation for the exact inference. Didn't find a good way to do it.

Despite their advantages, the posterior distribution in these models is often *doubly intractable*, making standard Markov Chain Monte Carlo (MCMC) methods challenging to apply. Current approaches rely on *approximate methods* (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021), which lack theoretical guarantees and explicit convergence bounds. This work addresses this gap by developing an *exact MCMC algorithm* (Bhattacharya, 2021; Murray; Ghahramani; MacKay, 2012) capable of efficiently sampling from these complex posterior distributions.

1.1 Formulation

Consider the following Bayesian schema:

$$p(\theta \mid D) = \frac{L(D \mid \theta)\pi_0(\theta)}{\int_\Theta L(D \mid t)\pi(t)\,\mathrm{d}t} = \frac{L(D \mid \theta)\ \overbrace{\pi_0(\theta)}^{\text{Initial prior}}}{m(D)}. \tag{1.1}$$

Definição 1.1.1 (Power prior)**.** *Let $D_0 = \{d_{01}, d_{02}, \ldots, d_{0N_0}\}$, $d_{0i} \subseteq \mathcal{X}^p$ be the historical data and let $L_0(D_0 \mid \theta)$ be a likelihood function assumed to be finite for all $\theta \in \Theta \subseteq \mathbb{R}^p$. Then, the power prior is defined as*

$$\pi_\eta(\theta \mid D_0) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta), \tag{1.2}$$

where $\pi_0(\theta)$ is the initial prior distribution of θ and η is the (fixed) power parameter.

Then, when we want to accomodate the uncertainty in the choice of η, we treat it as a random variable and assign a prior distribution to it, indexed by φ. In this case, we find the *unnormalized* power prior as

$$\pi^\star(\theta, \eta \mid D_0, \varphi) = \pi_\eta(\theta \mid D_0)\pi_A(\eta \mid \varphi) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta)\pi_A(\eta \mid \varphi). \tag{1.3}$$

In order to compute the posterior distribution of θ and η, we need to find the normalizing constant (from (1.2)) $c_0(\eta)$, which is given by

$$c_0(\eta) = \int_\Theta L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t) \tag{1.4}$$

and the marginal posterior distribution of η as, for a new dataset D,

$$p(\eta \mid D, D_0, \varphi) \propto \frac{\pi_A(\eta \mid \varphi)}{c_0(\eta)} \int_\Theta L(D \mid t) L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t). \tag{1.5}$$

Notice that, since both $c_0(\eta)$ and $\int_\Theta L(D \mid t)L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t)$ are intractable, we call this a doubly-intractable posterior. Previous work (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021) have focused on emulating $\log c_0(\eta)$ and $\log c'(\eta) = \mathbb{E}_{\pi_\eta}[\log L_0(D_0 \mid \theta)]$ through non-parametric models, which has proven to be more effective than common interpolation methods—particularly when incorporating shape constraints.

Sampling from these posteriors is challenging due to the intractability of standard M-H acceptance probabilities, making approximate methods like auxiliary variable techniques computationally expensive and lacking convergence guarantees; an alternative is unbiased event constructions, which enable exact sampling without evaluating normalizing constants.

For instance, applying Barker's algorithm for MCMC (Vats et al., 2021) to sample from a target distribution $\pi(x) \propto \pi'(x)$ with proposal density $q(x \mid y)$ has become feasible through Bernoulli factory methods. These methods enable the construction of events with probability $\alpha_B(x, y)$ when

$$\alpha_B(x, y) = \frac{\pi(y)q(y \mid x)}{\pi(x)q(x \mid y) + \pi(y)q(y \mid x)} = \frac{\pi'(y)q(y \mid x)}{\pi'(x)q(x \mid y) + \pi'(y)q(y \mid x)} \tag{1.6}$$

cannot be directly evaluated (Gonçalves; Łatuszyński; Roberts, 2017).

2 Background

3 Approximate inference with noisy-MCMC

Current approaches:

- Function emulation: (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021)

`Luiz:` (Han et al., 2023) (Carvalho; Ibrahim, 2021) and references therein

`Luiz:` Need to carefully read, cite and discuss (Park; Haran, 2020). Do their results hold for SCAM, for instance?

Our contributions in this area:

- Do MaL estimation and talk about GP and SCAM;
- Importance sampling to simplify and improve estimation of the NPP normalising constant (including control variates);
- Showcase on moderate size examples (including NIG linear regression as a baseline). Show that it works and then show how to break it

4 Exact inference for the Normalized Power Prior

- Estimators: Importance
- Scenarios:
- Models:

In this section, we develop exact Markov Chain Monte Carlo methods for inference with the normalized power prior (NPP). The key challenge lies in handling the intractable normalizing constant, which we address through Russian-roulette estimators that preserve the exactness of our sampling scheme.

4.1 Barker's Algorithm for Exact Inference

Barker's algorithm (Barker, 1965) provides an alternative to Metropolis-Hastings that is amenable to Bernoulli factory methods. For a target distribution $\pi(\theta)$ and proposal $q(\theta'|\theta)$, Barker's acceptance probability is:

$$\alpha_B(\theta, \theta') = \frac{\pi(\theta')q(\theta|\theta')}{\pi(\theta)q(\theta'|\theta) + \pi(\theta')q(\theta|\theta')} \tag{4.1}$$

While Barker's algorithm has lower efficiency than Metropolis-Hastings in the Peskun ordering, its acceptance probability has a crucial property: it can be expressed as a ratio of products. This enables the use of *Bernoulli factory* methods to implement the acceptance decision without explicitly computing α_B.

4.1.1 The Two-Coin Algorithm

The fundamental building block is the two-coin algorithm (Gonçalves; Łatuszyński; Roberts, 2017). Suppose we can factorize the acceptance odds as:

$$\frac{\pi(\theta')q(\theta|\theta')}{\pi(\theta)q(\theta'|\theta)} = \frac{c(\theta', \theta)p(\theta', \theta)}{c(\theta, \theta')p(\theta, \theta')} \tag{4.2}$$

where $c(\cdot, \cdot)$ is tractable and $p(\cdot, \cdot) \in [0, 1]$ is a probability we can simulate.

Then, Barker's acceptance probability becomes:

$$\alpha_B(\theta, \theta') = \frac{c(\theta', \theta)p(\theta', \theta)}{c(\theta', \theta)p(\theta', \theta) + c(\theta, \theta')p(\theta, \theta')} \tag{4.3}$$

The two-coin algorithm generates this probability without computing it explicitly:

Algorithm 1: Two-coin algorithm Just the idea

```
Flip coins with probabilities p(θ', θ) and p(θ, θ');
if both heads or both tails then
    return to step 1;
else if first heads and second tails then
    accept (return 1);
else if first tails and second heads then
    reject (return 0);
```

This procedure produces an acceptance event with exactly probability $\alpha_B(\theta, \theta')$.

4.2 Divide-and-Conquer Bernoulli Factory (DCBF)

For models with n observations, the likelihood factorizes as $\pi(\theta|D) \propto \prod_{i=1}^{n} f_i(\theta)$. The vanilla two-coin algorithm requires flipping a coin with probability $p(\theta) = \prod_{i=1}^{n} p_i(\theta)$. As n grows, $p(\theta)$ shrinks exponentially, causing the expected number of loops (ENL) to grow as $O(e^{cn})$ for some constant $c > 0$. This exponential cost is prohibitive even for moderate n

4.2.1 The Coin Merge Procedure

The DCBF addresses this by hierarchically merging factor coins. For two factors with odds h_1, h_2 and corresponding probabilities $r_j = h_j/(h_j + g_j)$, we can merge them into a single coin with probability $r_0 = h_0/(h_0 + g_0)$ where $h_0 = h_1 h_2$.

Algorithm 2: Coin merge algorithm Just the idea (again)

```
Flip coins with probabilities r_1 and r_2;
if both return the same value (both 0 or both 1) then
    return that value;
else
    return to step 1;
end
```

4.2.2 Hierarchical Factorization

Given n factors, we organize them in a balanced binary tree of depth $\ell = \lceil \log_2 n \rceil$.

At the leaves, we place individual factors (or small batches). At each internal node, we merge the coins from its two children.

Algorithm 3: Divide-and-Conquer Bernoulli Factory

```
function FlipDCBF({f_1, ..., f_n}, level k)
    if n = 1 then
        return Flip2Coin(f_1);
    end
    Split factors into {f_1, ..., f_{n/2}} and {f_{n/2+1}, ..., f_n};
    c_1 ← FlipDCBF(left half, k + 1);
    c_2 ← FlipDCBF(right half, k + 1);
    return MergeCoins(c_1, c_2);
end
```

4.2.3 Computational Complexity

(Stumpf-Fétizon; Gonçalves, 2025) prove that under regularity conditions, the expected cost of the DCBF scales as:

$$\mathbb{E}[\text{Cost}] = O(4^\ell \cdot \rho(n, \ell)) \tag{4.4}$$

where $\rho(n, \ell)$ is the cost of the two-coin algorithm at the leaves with batch size $n/2^\ell$.

For posterior concentration at rate $n^{-1/2}$ with optimal proposal scaling:

- If using likelihood ratio estimators: $\rho(n, \ell) = O(\sqrt{n/2^\ell})$
- Setting $4^\ell \propto n$ gives overall cost $O(n^{3/2})$

This is polynomial, compared to the exponential $O(e^{cn})$ cost of vanilla two-coin.

Moreover, this matches or improves upon pseudo-marginal algorithms, which typically require $O(n^2)$ cost.

4.3 Application to Normalized Power Prior

For the NPP, the joint posterior is:

$$\pi(\theta, \eta | D, D_0) \propto \frac{1}{c_0(\eta)} L(D|\theta) L_0(D_0|\theta)^\eta \pi_0(\theta) \pi_A(\eta) \tag{4.5}$$

The challenging term is the normalizing constant:

$$c_0(\eta) = \int_\Theta L_0(D_0|\theta)^\eta \pi_0(\theta) d\theta \tag{4.6}$$

4.3.1 Updating θ given η

`Adame:` Here I suggest substituting to regular MH

For updating θ, the full conditional is:

$$\pi(\theta|\eta, D, D_0) \propto L(D|\theta)L_0(D_0|\theta)^\eta \pi_0(\theta) \tag{4.7}$$

Since $L_0(D_0|\theta) = \prod_{i=1}^{N_0} f_i(\theta|D_{0i})$, we can use DCBF. For instance, when we have Gaussian data: $f_i(\theta|y_i) = \mathcal{N}(y_i|\theta, \Sigma)$.

The ratio form for Barker's algorithm is:

$$\frac{\pi(\theta'|\eta)}{\pi(\theta|\eta)} = \frac{L(D|\theta')}{L(D|\theta)} \cdot \frac{\pi_0(\theta')}{\pi_0(\theta)} \cdot \prod_{i=1}^{N_0} \left(\frac{f_i(\theta'|D_{0i})}{f_i(\theta|D_{0i})} \right)^\eta \tag{4.8}$$

The first two ratios are tractable. For the product term, we construct factor coins using likelihood ratios, which benefit from posterior concentration.

4.3.2 Updating η given θ

Updating η is more subtle because the full conditional involves $c_0(\eta)$:

$$\pi(\eta|\theta, D, D_0) \propto \frac{\pi_A(\eta)}{c_0(\eta)} L_0(D_0|\theta)^\eta \tag{4.9}$$

The ratio becomes:

$$\frac{\pi(\eta'|\theta)}{\pi(\eta|\theta)} = \frac{\pi_A(\eta')}{\pi_A(\eta)} \cdot \frac{c_0(\eta)}{c_0(\eta')} \cdot L_0(D_0|\theta)^{\eta'-\eta} \tag{4.10}$$

The term $c_0(\eta)/c_0(\eta')$ is intractable but can be estimated using

$$\frac{d}{d\tau} \log c_0(\tau) = \mathbb{E}_{\theta \sim \pi_\tau(\theta)} \left[\log L_0(D_0 \mid \theta) \right], \tag{4.11}$$

where $\pi_\tau(\theta) \propto L_0(D_0 \mid \theta)^\tau \pi_0(\theta)$ is the power posterior at temperature τ.

Integrating this identity from η to η' yields the log-ratio:

$$\Delta = \log \frac{c_0(\eta')}{c_0(\eta)} = \int_\eta^{\eta'} \mathbb{E}_{\theta \sim \pi_\tau(\theta)} \left[\log L_0(D_0 \mid \theta) \right] \, \mathrm{d}\tau \,. \tag{4.12}$$

4.3.3 Practical implementation

In practice, we replace the integral Δ with a simple Monte Carlo estimation. This is a unbiased estimator for the actual integral:

$$\widehat{\Delta}_a(\eta,\eta') = \frac{(\eta'-\eta)}{K}\sum_{k=1}^{K}\mathbb{E}_{\theta\sim\pi_{\tau_k}(\theta)}\left[\log L_0(D_0 \mid \theta)\right], \text{ and so} \tag{4.13}$$

$$\widehat{\Delta}_b(\eta,\eta') = \frac{(\eta'-\eta)}{M\cdot K}\sum_{k=1}^{K}\sum_{m=1}^{M}\log L_0\Big(D_0 \mid \widetilde{\theta}_{m_k}\Big), \tag{4.14}$$

where $\tau_k \sim \text{Uniform}(\eta,\eta')$, and $\left(\widetilde{\theta}_{m_k} \mid \tau_k\right) \sim \pi_{\tau_k}(\theta)$.

Thus, for a single sample $\tau_\star \sim \text{Uniform}(\eta,\eta')$, and $\left(\widetilde{\theta}_\star \mid \tau_\star\right) \sim \pi_{\tau_\star}(\theta)$ we have:`Adame:` Because it doesn't depend on the variance

$$\widehat{\Delta}(\eta,\eta') = (\eta'-\eta)\log L_0\Big(D_0 \mid \widetilde{\theta}_\star\Big) \tag{4.15}$$

4.3.4 Producing an unbiased estimator via Poisson Estimator

As discussed in (Stumpf-Fétizon; Gonçalves, 2025), we can produce an unbiased estimator $\widehat{I}$ such that $\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] = \exp\{-\Delta\}$, where $K \sim \text{Poisson}(\lambda)$ and $(\tau_\star,\theta_\star)_j \sim P$ (built from the conditional and marginal). In fact, we sample $\tau_{\star j} \sim \text{Uniform}[\eta,\eta']$. `Adame:` Note here, we have to define this measure properly. It should be a product measure.

In that case, we find

$$\widehat{I} = \exp\{(\lambda-c)\}\prod_{j=1}^{K}\frac{(c-\widehat{\Delta}(\tau_{\star j}))}{\lambda} \tag{4.16}$$

The choice of λ and c is crucial for the efficiency of the estimator. An optimal choice would be to take $\lambda = \sup_{\theta\in\Theta}\widehat{\Delta} - \inf_{\theta\in\Theta}\widehat{\Delta}$ and $c = \sup_{\theta\in\Theta}\widehat{\Delta}$. Refer to seção A.1 for a detailed proof of unbiasedness.

In order to make proper use of this method with $\widehat{\Delta}(\eta,\eta')$, one would use a prior with finite support. We then suggest employing a standard-normal truncated at the 99% percentiles.

`Adame:` Add the new estimators

4.4 Experiments

4.4.1 Building an estimator

The most simple approach is the harmonic estimator, since

$$c_0(\eta) = \mathbb{E}_{\theta\sim\pi_0}[L_0(\theta \mid D_0)^\eta].$$

Then, we define δ_a as

$$\delta_a(\eta, \theta_1, \ldots, \theta_K) = \frac{1}{K} \sum_{i=1}^{K} L_0(\theta_i \mid D_0)^\eta \tag{4.17}$$

And with this estimator we can then employ a Portkey Barker's (Vats et al., 2021) to sample from the power posterior.

This δ_a has a major two-sided feature, while it is easy to sample from the initial prior, its efficiency is low due to its large non-informativeness.

In order to improve the efficiency, we can produce an importance sampling estimator by concentrating the probability density around the $\hat{\theta}_{MLE}$, which gives us

$$c_0(\eta) = \int_{\theta \in \Theta} L_0(\theta \mid D_0)^\eta \pi_0(\theta)\, \mathrm{d}\theta = \mathbb{E}_{\theta \sim \pi_{MLE}} \left[L_0(\theta \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{MLE}(\theta)} \right].$$

We can then define δ_b as:

$$\delta_b(\eta, \theta_1, \ldots, \theta_K) = \frac{1}{K} \sum_{i=1}^{K} L_0(\theta_i \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{MLE}(\theta)}. \tag{4.18}$$

`Adame:` Different compatibility scenarios.

4.4.2 The choice of λ and c for the Gaussian likelihood with known Σ.

Considering the functional form of our estimator $\widehat{\Delta}(\eta, \eta') = (\eta' - \eta) \log L_0\left(D_0 \mid \widetilde{\theta}\right)$ and that L_0 is the likelihood of $D_0 = \{d_{01}, \ldots, d_{0N}\}$ where each $d_{0j} \sim \mathcal{N}_d(\theta, \Sigma)$ for $j = 1, \ldots, N$ with known Σ, we can find that

$$\arg\max_{\theta \in \Theta} \log L_0(D_0 \mid \theta) = \hat{\theta}_{\mathrm{MLE}} = \frac{1}{N} \sum_{j=1}^{N} d_{0j},$$

and, therefore, we have that

$$\max_{\theta \in \Theta} \log L_0(D_0 \mid \theta) = -\frac{Nd}{2} \log(2\pi) - \frac{N}{2} \log \det \Sigma - \frac{N}{2} \operatorname{tr}(\Sigma^{-1} S),$$

where $S = \frac{1}{N} \sum_{j=1}^{N} (x_j - \hat{\theta}_{\mathrm{MLE}})(x_j - \hat{\theta}_{\mathrm{MLE}})^\top$

4.4.3 The choice of λ and c for the exponential family

`Adame:` Produce analytical results for exponential family.

A canonical exponential family has the following density:

$$L(X \mid \theta) = b(X) \exp\Big\{\theta^\top t(X) - c(\theta)\Big\},\ \theta \in \Theta \subseteq \mathbb{R}^d, \tag{4.19}$$

where Θ is open and connected.

5 Experiments

6 Discussion

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APPENDIX A – Proofs

A.1 Proof that 4.16 is unbiased

Demonstração. The result is immediate, we first take the expected value with respect to P that leaves us with the actual Δ inside the expected value.

$$\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] = \mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\exp\{(\lambda - c)\}\prod_{j=1}^{K}\frac{(c-\widehat{\Delta}(\tau_{\star j}))}{\lambda}\right], \tag{A.1}$$

$$= \mathbb{E}_K\left[\exp\{(\lambda - c)\}\prod_{j=1}^{K}\frac{(c-\Delta)}{\lambda}\right], \tag{A.2}$$

$$= \mathbb{E}_K\left[\exp\{(\lambda - c)\}\left(\frac{(c-\Delta)}{\lambda}\right)^K\right], \tag{A.3}$$

then, we take the expected value with respect to $K \sim \text{Poisson}(\lambda)$

$$\begin{aligned}
\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] &= \sum_{i=0}^{\infty}\mathbb{P}[K=i]\exp\{\lambda - c\}\left(\frac{(c-\Delta)}{\lambda}\right)^i, \\
&= \sum_{i=0}^{\infty}\frac{e^{-\lambda}\lambda^i}{i!}\exp\{\lambda - c\}\left(\frac{(c-\Delta)}{\lambda}\right)^i, \\
&= e^{-c}\sum_{i=0}^{\infty}\frac{(c-\Delta)}{i!} = e^{-c}e^{c-\Delta} = e^{-\Delta}.
\end{aligned}$$

□